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SUPERSEDING
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8 December 1972

MILITARY SPECIFICATION
COMPOUND, INSULATING, POTTING AND ENCAPSULATING, EPOXY RESIN

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of two part epoxy resin compound suitable for molding, potting, and encapsulating applications. The compounds are supplied in two part kits of epoxy resin and hardener. When cured, the compounds become a rigid impact and thermal shock resistant material capable of mechanically supporting embedded parts or assemblies.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials

ASTM D 150	A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials, Methods of Test for
ASTM D 256	Impact Resistance of Plastics and Electrical Insulating Materials, Methods of Test for
ASTM D 257	D-C Resistance or Conductance of Insulating Materials, Methods of Test for
ASTM D 570	Water Absorption of Plastics, Method of Test for
ASTM D 638	Tensile Properties of Plastics, Method of Test for
ASTM D 648	Deflection Temperature of Plastics Under Load, Method of Test for
ASTM D 757	Flammability of Plastics, Self-Extinguishing Type, Method of Test for
ASTM D 790	Flexural Properties of Plastics, Method of Test for
ASTM D 1763	Epoxy Resins, Specification for
ASTM D 1924	Determining Resistance of Plastics to Fungi, Recommended Practice for
ASTM D 2240	Indentation Hardness of Rubber and Plastics by Means of a Durometer, Method of Test for

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ASTM D 2393 Viscosity of Epoxy Compounds and Related Components, Method of Test for

ASTM D 2566 Linear Shrinkage of Thermosetting Casting Systems During Cure, Method of Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60605.)

National Motor Freight Traffic Associations, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street N.W., Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall be required and shall meet all the requirements of this specification.

3.2 Material.

3.2.1 Filled resin base. Unfilled resin base material shall meet the requirements of ASTM D 1763, Type I, Grade 1, for epoxy resin having a viscosity of 11,000 to 18,000 centipoises (cps) at plus 25 degrees Celsius (C) plus or minus 2 degrees C (plus 77 degrees Fahrenheit (F) plus or minus 4 degrees F). The filled resin system shall contain 50 percent plus or minus 2 percent by weight of quartz silica filler and shall have a viscosity of 60,000 to 100,000 cps at plus 25 plus or minus 2 degrees C (plus 77 plus or minus 4 degrees F).

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3.2.2 Hardener. The hardener shall be a polyfunctional aliphatic amine having a viscosity of 14 to 25 cps at plus 25 plus or minus 2 degrees C (plus 77 plus or minus 4 degrees F).

3.2.3 Colored paste. Colored paste may be added to the filled resin base. The color of the material is optional unless otherwise specified in the contract or order.

3.3 Compound requirements. The compound shall conform with the following requirements when the filled resin base and hardener are mixed in accordance with the manufacturer's instructions.

WARNING

Avoid contact with skin and eyes and use with good ventilation to draw fumes away from operator personnel.

3.3.1 Work life. The working life shall be a minimum of 30 minutes at plus 26 plus or minus 5 degrees C (plus 80 plus or minus 10 degrees F). Working life is the time during which the compound will flow when its container is tipped at an angle no greater than 45 degrees. The height of the material in the container shall be equal to its diameter.

3.3.2 Cure time. The cure time to attain a Shore "D" hardness per Table I shall be in accordance with each of the following schedules:

a. Twenty-four hours maximum at plus 26 plus or minus 5 degrees C (plus 80 plus or minus 10 degrees F).

b. Two hours maximum at plus 66 plus or minus 5 degrees C (plus 150 plus or minus 10 degrees F).

3.4 Cured material requirements. The materials shall meet the requirements of Table I when mixed in accordance with the manufacturer's instructions and cured in accordance with 3.3.2.

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Table I. Requirements of cured materials

Property	Requirement
Tensile strength (psi)	8,000
Flexural strength (psi)	12,000
Hardness (Shore D)	80 ± 5
Izod impact strength (ft - lb/in. of notch)	0.3 minimum
Heat deflection temperature (°C at 264 psi)	60

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Table I. Requirements of cured materials (continued)

Property	Requirement
Linear shrinkage (%)	0.55
Moisture absorption (weight % for 24 hour immersion)	0.25
Dielectric constant at plus 30°C (plus 86°F)	4.21
Volume resistivity at plus 30°C (plus 86°F) (ohm - cm)	3.99×10^{14}
Flammability	Self-extinguishing
Fungus resistance	non-nutrient

3.5 Shelf life. The shelf life of the individual components shall be 12 months from date of manufacture, when stored in airtight containers under warehouse storage conditions of plus 16 to plus 38 degrees C (plus 60 to plus 100 degrees F) in the absence of sunlight.

3.6. Workmanship. The individual components shall be uniform in quality and free from impurities and other defects which could adversely affect their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless diapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods for the preparation of subsequent units of product. Preproduction samples which

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do not meet all the requirements of this specification shall be subject to rejection and returned to the supplier.

4.3 Classification of tests. The testing of the material shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance inspection. Quality conformance inspection for the acceptance of the material shall consist of the following examinations and tests:

a. Visual.

b. Viscosity of filled resin and hardener.

c. Hardness (use either cure schedule of 3.3.2).

4.3.2.1 Lot size. Lot size shall consist of all the materials manufactured at one time, from one batch, forming a part of one contract or order, and submitted for acceptance at one time and place.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), one pint of the filled resin and one pint of the hardener, representative of each lot shall be selected at random for quality conformance inspection. Each container of resin and hardener shall be considered as a unit of product. Failure of the samples to meet quality conformance requirements shall be cause for lot rejection.

4.4 Test methods. Test methods shall be as follows.

4.4.1 Visual examination. A visual examination of the material shall be conducted to determine compliance with 3.6 as well as the preservation, packaging, packing and marking of Section 5.

4.4.2 Viscosity. Viscosity of the filled resin base and the hardener shall separately be determined in accordance with ASTM D 2393 to show compliance with 3.2.1 and 3.2.2.

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4.4.3 Color. The color shall be visually examined to determine compliance with 3.2.3.

4.4.4 Work life. Thoroughly mix two hundred grams of the filled base resin with an amount of hardener in accordance with the manufacturer's instructions, at plus 26 plus or minus 5 degrees C (plus 80 plus or minus 10 degrees F). The container shall be cylindrical and of such size that the height of the material shall be equal to its diameter. The work life shall be considered at an end when the material will no longer flow when the container is tipped at an angle of 45 degrees.

4.4.5 Cure time. Material shall be cured to each of the cure schedules of 3.3.2 and tested to meet the requirements of 3.4.

4.4.6 Tensile strength. Tensile strength of the cured material shall be determined in accordance with ASTM D 638 and conform to 3.4.

4.4.7 Flexural strength. Flexural strength of the cured material shall be determined in accordance with ASTM D 790 and conform to 3.4.

4.4.8 Hardness. Hardness of the cured material shall be determined in accordance with ASTM D 2240 and conform to 3.4.

4.4.9 Izod impact strength. Izod impact strength of the cured material shall be determined in accordance with ASTM D 256 and conform to 3.4.

4.4.10 Heat deflection temperature. Heat deflection temperature of the cured material shall be determined in accordance with ASTM D 648 and conform to 3.4.

4.4.11 Linear shrinkage. Linear shrinkage of the cured material shall be determined with ASTM D 2566 and conform to 3.4.

4.4.12 Moisture absorption. Moisture absorption of the cured material shall be determined in accordance with ASTM D 570 and conform to 3.4.

4.4.13 Dielectric constant. Dielectric constant of the cured material shall be determined in accordance with ASTM D 150 and conform to 3.4.

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4.4.14 Volume resistivity. Volume resistivity of the cured material shall be determined in accordance with ASTM D 257 at plus 30 degrees C (plus 86 degrees F).

4.4.15 Flammability. Flammability of the cured material shall be determined in accordance with ASTM D 757 and conform to 3.4.

4.4.16 Fungus resistance. Fungus resistance of the cured material shall be determined in accordance with ASTM D 1924 and conform to 3.4.

4.5 Shelf life. Periodic inspections shall be conducted to insure conformance with 3.5 and proper methods shall be initiated for the disposal of outdated products.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. The filled resin and the hardener shall be packaged in separate suitable containers as a unit package. A unit package shall consist of one container of filled resin and one container of hardener which shall be adequately packaged to protect the contents. In addition, at least one instruction sheet shall be included in each unit package. The instructions sheet shall contain as a minimum the following information:

a. Mixing instructions.
b. Time and temperature for cure cycle.
c. Necessary safety and toxicity information and precautions to be observed during all operations in using the sealant.
d. Any other information relative to the use and storage of the product.

5.2 Packing.

5.2.1 Level A. Unit packages of intermediate containers packaged as prescribed herein shall be packed in a container that will insure safe delivery at destination and in compliance with Uniform Freight Classification or other carrier regulations applicable to the mode of transportation.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following:

a. Name and address of manufacturer.

b. Quantity of material or net weight.

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c. Date of manufacture and batch or lot number.

d. Deterioration and storage life instructions.

e. Contract or purchase order number.

6. NOTES

6.1 Intended use. The material covered by this specification, is intended for molding, potting, and encapsulating applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required, and if so, pertinent details.

c. Sampling plan if other than specified (see 4.3.2).

d. Quantity and size of containers.

e. Color of material when a particular color is required (see 3.2.3).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification, MIS-13516, dated 8 December 1972.

Custodian:
Army-MI

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